

Tutorial No : 6

Tutorial Title: Write a program in C++ to perform following operations on complex numbers Add, Subtract, Multiply, Divide. Use operator overloading for these operations. The objective of this assignment is to learn the concepts of Operator overloading.

Name of Student: Vikiraj Vadghane

Class: SY

Div: 2

PRN No: B25ET1200

Batch: C

Program:

```
#include <iostream>
using namespace std;

class complex {
    float real, img;

public:
    void accept(float r, float i) {
        real = r;
        img = i;
    }

    void display() {
        if (img >= 0) {
            cout << real << " + " << img << "i";
        } else {
            cout << real << " - " << -img << "i";
        }
    }

    complex operator+(complex c2) {
        complex sum;
        sum.real = real + c2.real;
        sum.img = img + c2.img;
        return sum;
    }

    complex operator-(complex c2) {
        complex sub;
```

```
    sub.real = real - c2.real;
    sub.img = img - c2.img;
    return sub;
}
```

```
complex operator/(complex c2) {
    complex multiply;
    multiply.real = real * c2.real;
    multiply.img = img * c2.img;
    return multiply;
}
```

```
complex operator*(complex c2) {
    complex division;
    division.real = real / c2.real;
    division.img = img / c2.img;
    return division;
}
};
```

```
int main() {
    complex c1, c2, c3, c4, c5, c6;
    c1.accept(3, -5);
    c2.accept(9, 20);

    cout << "c1 = ";
    c1.display();
    cout << "\nc2 = ";
    c2.display();
    cout << endl;

    c3 = c1 + c2;
    cout << "the addition of two complex: ";
    c3.display();
    cout << endl;

    c4 = c1 - c2;
    cout << "the subtraction of two complex: ";
    c4.display();
    cout << endl;
}
```

```
c5 = c1 / c2;  
cout << "the multiplication of two complex: ";  
c5.display();  
cout << endl;  
  
c6 = c1 * c2;  
cout << "the division of two complex: ";  
c6.display();  
cout << endl;  
  
return 0;  
}
```

Output:

```
c1 = 3 - 5i  
c2 = 9 + 20i  
the addition of two complex: 12 + 15i  
the subtraction of two complex: -6 - 25i  
the multiplication of two complex: 27 - 100i  
the division of two complex: 0.333333 - 0.25i
```

```
=== Code Execution Successful ===
```